Catchment – Aware Flood Forecasting System

Perera B.T.S
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21819360@my.sliit.lk

Amarasekara S.K
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21340246@my.sliit.lk

Dias A.H.S.G
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21305214@my.sliit.lk

Samaranayake N.A.S.R
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21816772@my.sliit.lk

Narmada Gamage
Department of Computer Systems Engineering, Faculty of Computing, Sri Lanka Institute of Information Technology, Malabe, Sri Lanka
narmada.g@sliit.lk

Pradeep Abeygunawardhana
Department of Computer Systems Engineering, Faculty of Computing, Sri Lanka Institute of Information Technology, Malabe, Sri Lanka
pradeep.a@sliit.lk

Abstract—

Keywords— RF*(Random Forest)****,SVM(****Support Vector Machine****),ANN(****Artificial Neural Network****), HEC-RAS(****Hydrologic Engineering Center's River Analysis System****), MQTT(**Message Queuing Telemetry Transport**), XAI** (Explainable Artificial Intelligence), **LSTM** (Long Short-Term Memory), **UI** (User Interface), **NGO**(Non-Governmental Organization), **API**(Application Programming Interface)*

I. Introduction

Flooding remains one of the most prevalent and destructive natural disasters in Sri Lanka, primarily caused by excessive rainfall, inadequate drainage infrastructure, and rapid water level rises in river catchments. Although platforms such as RiverNet.lk provide real-time water level data, their effectiveness is limited by significant transmission delays and latency in data updates, which hinders timely flood warnings and efficient emergency response [1].

To address these limitations, this research proposes a Catchment-Aware Flood Forecasting System that integrates Edge Computing, the Internet of Things (IoT), and Explainable Artificial Intelligence (XAI). Unlike traditional systems that rely heavily on centralized cloud computing, this system utilizes edge devices to perform localized, real-time analysis of environmental data. This reduces latency and enhances the speed and accuracy of flood risk predictions [2].

IoT sensors deployed in critical river catchments will collect real-time hydrological and meteorological data—such as rainfall and water levels—which are processed immediately at the edge. This local data handling ensures quicker insights and reduces dependence on external systems. In parallel, Explainable AI models provide transparent, human-understandable flood predictions that help stakeholders—including local communities, emergency responders, and policymakers—grasp the rationale behind forecasts and take timely action [2] [3].

Moreover, the proposed solution includes mobile and web-based applications to deliver real-time alerts, visualize flooded zones, suggest alternate routes, and provide guidance to nearby safe locations and essential resources. These features aim to empower users with accurate, actionable information, enhancing both preparedness and responsiveness.

Ultimately, this study seeks to establish a scalable, resilient, and technologically advanced system to transform flood forecasting in Sri Lanka, minimizing the adverse impacts of floods and strengthening national disaster management capabilities.

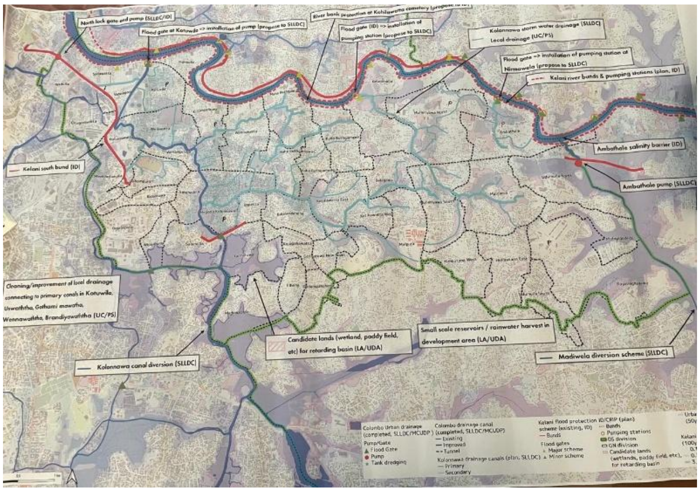

Figure 1 Flood Hazard Map in the Kolonnawa Area

II. Literature review

Reducing the risk of disasters depends on accurate flood forecasts, particularly in areas like Sri Lanka that are vulnerable to flooding. By incorporating contemporary technologies like cloud computing, artificial intelligence (AI), and the Internet of Things (IoT) into early warning systems, numerous studies have tackled this problem. Latency, transparency, and actionable prediction issues are still present, though.

A. Flood Impact Prediction and Optimization-Driven Disaster Response Systems

Flood disaster management increasingly leverages machine learning (ML) and optimization to enhance predictive accuracy and operational response. Among ML approaches, Random Forest (RF) consistently outperforms others in flood impact estimation due to its ability to model complex, non-linear relationships and robustness to limited or noisy data [28][29]. For instance, RF has been successfully used to generate flood hazard susceptibility maps in diverse regions, often surpassing the performance of single-model or rule-based approaches [29].

However, for flood prediction outcomes to be trusted and actionable, transparency is crucial. Explainable AI (XAI) frameworks, which highlight feature influence and model behavior, have been applied to flood systems to improve stakeholder trust [30] [31]. Integrating XAI with RF ensures that users can understand the rationale behind predictions, making such systems operationally reliable in disaster scenarios.

Beyond prediction, resource optimization plays a central role in flood emergency response. Studies in humanitarian logistics emphasize that multi-objective optimization models, including linear programming and genetic algorithms, are highly effective for allocating scarce resources like food, shelters, and rescue teams [32] [34]. These techniques help ensure fair, need-based distribution even under strict supply and time constraints.

Moreover, recent research blends fuzzy logic and stochastic programming to manage uncertainty in post-disaster logistics [35] [36]. Such adaptive models are particularly relevant in flood contexts, where ground conditions and supply availability change dynamically.

This research integrates these methodologies into a unified system

I. A Random Forest model predicts flood impact metrics (people affected, families impacted, houses destroyed) using meteorological and hydrological inputs.
II. An optimization algorithm based on linear programming allocates resources efficiently based on predicted severity and population impact.
III. A language generation module produces multilingual situation reports (Sinhala, Tamil, English), improving accessibility and real-time response readiness.

By combining interpretable ML, optimization, and localized communication, this component supports faster, fairer, and more inclusive flood response in vulnerable areas like the Kelani River Basin.

B. Real-Time Flood risk prediction, visualization, and Safe Route Identification for Transportation.

Recent advances in machine learning-based flood prediction systems demonstrate Random Forest (RF) algorithms as the optimal choice for multi-level flood risk classification, achieving 85-98% accuracy rates across diverse riverine environments [4], [5]. Studies specifically implementing three-tier risk classification (low, medium, high) report 77-95% accuracy, with RF outperforming SVM (83-90%) and ANN (85-92%) alternatives [6]. The integration of IoT sensor networks utilizing HC-SR04 ultrasonic sensors provides real-time water level monitoring [7]. Modern flood visualization systems adopt standardized color-coded risk levels following international emergency communication protocols, with web-based platforms achieving 30-minute update intervals for real-time flood extent mapping [8], [9]. Safe route identification algorithms employing modified Dijkstra's algorithm with flood-aware weighting demonstrate 0.012-second processing times, though evacuation paths may increase by 33.17% during extreme flood events [10], [11]. Kaduwela and Megoda Kolonnawa areas experience 6-10 feet flood depths, with 85% of events occurring during June-September monsoons, necessitating robust real-time monitoring systems [12]. Climate change projections indicating a 16% increase in design flood volumes and 51% urbanization growth in the Colombo region (1997-2015) underscore the urgency of implementing advanced flood prediction systems that combine machine learning accuracy with real-time sensor data for effective risk communication and transportation safety management [13].

C. IOT device management and Dashboard

Surge observing frameworks depends on a combination of vigorous innovations to supply exact, convenient, and solid cautions. Waterproof ultrasonic sensors are fundamental as they can persevere unforgiving climate and provide exact water level readings, enduring 40–60% longer than standard sensors. The ESP8266 microcontroller acts as the system's brain, advertising quick information preparing, moo control utilization, and built-in Wi-Fi for productive communication. For zones missing dependable web, the SIM900A module empowers cellular network, permitting opportune SMS alarms with up to 98% victory indeed in country districts. GPS modules improve area precision, pinpointing surge zones inside 3–5 meters, which makes a difference crisis groups react 25–35% quicker. A neighborhood LCD show guarantees that real-time data remains obvious amid blackouts, expanding open consideration to notices by up to 40%. When combined, these components shape a self-sustained framework that switches between Wi-Fi and portable systems for 99% communication uptime, forms information locally for speedier cautions, and can work for months or indeed uncertainly with sun-oriented control and productive vitality use. Housed in weatherproof walled in areas that withstand extraordinary conditions, the framework requires negligible support and keeps going 3–5 a long time. At long last, centralized IoT dashboards permit administrators to remotely oversee gadgets, screen sensor wellbeing, visualize surge information on intelligently maps, and react speedier utilizing prescient analytics—making the whole setup a keen, versatile, and life-saving arrangement for present day surge hazard administration.

D. Provide security guidance for flood-affected individuals to find safe sites and essential supplies.

Flooding continues to be a persistent and destructive natural disaster in Sri Lanka, particularly in the Colombo District, where low-lying urban areas like Megoda Kolonnawa and Kaduwela experience 6–10 feet of flood depths, primarily during the June–September monsoon season. Traditional flood forecasting systems often fail to provide timely, actionable guidance for vulnerable populations. To overcome these challenges, there has been a significant shift toward intelligent flood management systems that integrate real-time monitoring, AI-driven decision-making, and IoT-based sensing technologies.

Recent literature underscores the importance of predictive analytics in flood response planning. Machine Learning (ML) models, including LSTM networks, have shown strong performance in forecasting water levels and flood risks based on historical and real-time hydrological data [14].These models are especially suited to catchment-sensitive environments like the Kelani River basin, which impacts areas such as Megoda Kolonnawa. By incorporating Explainable AI (XAI) methods like SHAP and LIME, flood prediction outputs can be made interpretable to both emergency responders and affected individuals, increasing system trust and compliance [14]

Geospatial mapping and real-time user tracking also play a pivotal role in ensuring effective evacuation and supply distribution. GIS-based flood visualization, when combined with mobile tracking, can drastically reduce response times and guide users to safer zones during rising floodwaters [15]. Importantly, your component adds a novel contribution by predicting the required quantities of essential items like food and medicine for specific safe sites, using trained models on past disaster data. This addresses a gap identified in recent studies, which highlight the logistical challenges in resource planning during floods [16].

The deployment of IoT sensors enhances system responsiveness by providing continuous updates on environmental metrics such as rainfall, river levels, and localized flooding [17]. This real-time sensing is crucial in Sri Lanka, where existing systems such as RiverNet.lk suffer from update latency. Moreover, two-way communication mechanisms improve adaptability by allowing affected individuals to request resources or report urgent needs, further personalizing disaster response [18].

Collectively, these technologies create a robust, user-centric framework for flood risk mitigation. Your system's emphasis on data-driven essential item allocation, real-time risk visualization, and interpretable AI navigation makes it well-suited to support disaster resilience in flood-prone Sri Lankan urban zones.

III. METHODOLOGY

This research employs a comprehensive approach to develop an IoT-integrated, edge-computing-based, and Explainable AI-driven flood forecasting system specifically designed for the Kelani River in Sri Lanka. The research adopts a multi-component, mixed-methods framework that addresses the complex challenges of real-time flood prediction in urban riverine environments. The methodology integrates advanced sensor technologies, edge computing architectures, machine learning algorithms, and explainable AI techniques to create a robust, scalable, and transparent flood forecasting system. As part of this system, a dedicated component was developed to predict the potential impact of flooding—including the number of people, families, and houses affected—using a Random Forest model. The predicted outputs are then used to dynamically allocate limited emergency resources through a linear programming-based optimization algorithm. Additionally, the system generates multilingual situation reports and safety recommendations in Sinhala and Tamil, enhancing communication and accessibility for local communities. This approach is justified by the need for high-resolution, real-time monitoring capabilities combined with reliable communication infrastructure and interpretable prediction models that can support informed decision-making by disaster management authorities and affected communities.

A. Data Collection: Sources

This research employs a multi-source data collection approach to develop a comprehensive IoT-integrated, edge-computing-based, and Explainable AI-driven Catchment Aware flood forecasting. The methodology integrates real-time sensor data, historical hydrological records, and meteorological information to create a robust prediction framework. The primary data collection focuses on real-time hydrological parameters from strategically positioned IoT sensor nodes across the Kaduwela, Megoda Kolonnawa river catchment of the Kelani river. These catchments were selected based on their critical importance to population centers and historical flood vulnerability patterns. The sensor network captures water level measurements at 5-15 minute intervals, following the real-time monitoring standards established for riverine flood detection systems [19]. Historical flood records from the Department of Irrigation, Sri Lanka, provide crucial baseline data. These records include water level measurements, flood extent maps, and damage assessments that enable model training and validation. The Disaster Management Centre of Sri Lanka contributes standardized flood risk classifications (Alert, Minor, Major, Critical levels) that inform the system's alert thresholds and decision-making protocols.

B. Data Collection Methods

The data collection methodology implements a three-tier IoT architecture that ensures comprehensive coverage while maintaining system resilience. This approach builds upon established frameworks for real-time flood monitoring, adapted for Sri Lankan hydrological conditions [20]. Ultrasonic water level sensors are deployed at critical monitoring points identified through hydrological analysis and historical flood patterns. Each sensor node consists of an ultrasonic sensor module connected to an ESP8266 microcontroller, providing 1cm accuracy for water level measurements as validated in IoT-based flood monitoring systems [21]. The sensors are positioned above maximum historical flood levels and protected by weatherproof enclosures. The edge computing layer processes raw sensor data locally before transmission, implementing established architectures for real-time flood monitoring [22]. The LC display module provides local visualization for mainstream personnel and community observers, fostering stakeholder engagement in the monitoring process. Data transmission utilizes the MQTT (Message Queuing Telemetry Transport) protocol over cellular networks via SIM900A GSM modules, ensuring reliable communication even in remote locations. The publish-subscribe architecture of MQTT enables efficient many-to-many communication between sensor nodes and processing servers, as demonstrated in environmental monitoring applications [23].

C. Data Analysis Methods and Techniques

The analytical framework combines edge computing capabilities with cloud-based machine learning to deliver real-time, explainable flood predictions. The IoT architecture employs Azure IoT Hub as the central communication backbone, where ESP8266 sensors transmit water level data via MQTT protocol to local Azure IoT Edge gateways. Azure IoT Edge modules process this telemetry data locally using containerized anomaly detection services, applying machine learning models trained on historical flood patterns. When anomalies are detected, the system triggers immediate local alerts while simultaneously sending critical data to Azure IoT Hub for cloud-based analysis and dashboard visualization. The IoT solution leverages Azure Stream Analytics at the edge to perform real-time data filtering and aggregation, reducing bandwidth usage by transmitting only relevant anomaly events rather than continuous sensor streams. Device twins in Azure IoT Hub enable remote configuration of detection thresholds and algorithm parameters across the distributed sensor network without physical access to field devices. This IoT implementation ensures scalable deployment across multiple watersheds, with centralized monitoring through Azure IoT Central dashboards and automated alert distribution to emergency response teams via Azure Logic Apps. The cloud-based analytics platform employs an ensemble approach combining Long Short-Term Memory (LSTM) networks with Random Forest algorithms, implemented using TensorFlow and Scikit-learn respectively. This combination leverages LSTM's superior performance for time-series water level forecasting, achieving 68.5-76.1% accuracy within required precision intervals as demonstrated in recent hydrological studies [24], while Random Forest provides robust flood risk classification with up to 99.8% accuracy reported in comparative studies [25]. To ensure stakeholder trust and facilitate informed decision-making, the system implements SHAP (Shapley Additive exPlanations) for model interpretation, following methodologies established for geohazard prediction [26].

D. Tools and Materials Used

Components	Technology and Tools
Hardware Components	ultrasonic sensors (HC-SR04 or waterproof JSN-SR04T variants) for water level measurement, ESP8266 NodeMCU microcontrollers serve as the edge computing platform, offering WiFi connectivity, sufficient processing power for edge analytics, and low power consumption suitable for solar-powered operation.
Software Development Environment	Visual Studio Code serves as the primary integrated development environment
Backend Infrastructure	MySQL for structured time-series sensor data, with optimized schemas for efficient temporal queries Firebase Realtime Database complements MySQL by providing instant synchronization for alert notifications and mobile application integration
Machine Learning Framework	TensorFlow 2.x implements the LSTM networks for time-series forecasting, leveraging GPU acceleration for model training and inference Scikit-learn provides the Random Forest implementation and supporting utilities for data preprocessing, feature engineering, and model evaluation.
Deployment and Monitoring	Firebase Cloud Messaging enables push notifications to mobile applications, ensuring timely alert delivery to stakeholders and affected communities AWS CloudWatch monitors system performance, tracking metrics such as prediction latency, model accuracy drift, and sensor network health.

E. Visualization of Results

The results of the system were visualized through several methods for the better understandability of the community users and the authorities. Visualizing flooded and predicted flood-affected locations on the map and graphical visualization of the three risk stages (high risk, medium risk, low risk) using color codes enhanced understandability. Identified alternative routes through the algorithm and the flood risk predictions are suggested and guide community users by navigating them through Google Maps in order to pass through areas safely, to reach their destinations.

The system's findings were represented using intuitive and interactive interfaces to help community members and emergency response authorities easily evaluate crucial safety information. Safe places were dynamically depicted on a map, allowing users to see their proximity, status, and availability of critical supplies such as food, water, and medicine. These safe places were color-coded to show their capacity levels (e.g., green for available, yellow for nearly full, and red for full), allowing users to make informed evacuation decisions. Furthermore, the amount of vital supplies projected and allocated at each site was shown using bar graphs, which provided insights into distribution patterns and demand against supply across different regions. The system also displayed heatmaps showing population concentration at shelters and places with the greatest supply needs.

In addition to spatial and resource visualization, the system displayed predicted damage levels—including affected families, people, and houses—using area-wise summary cards and severity indicators. These predictions, generated through a Random Forest model, were linked to the resource allocation module, which dynamically calculated and displayed the distribution of limited resources such as boats, shelters, and food packs. Visual dashboards highlighted the real-time allocation results, severity priority levels, and remaining supplies available for redistribution.

Navigation information to these secure sites has been integrated into Google Maps, allowing users to take the safest path in real time. Alert banners and push alerts visually displayed the safest local location, as well as the projected trip time, potential barriers, and updated resource availability.

For authorities, a dashboard was created that displays the user input, resource request trends, system-predicted impact values, and site overcrowding alarms, allowing for more effective resource reallocation and decision-making during floods. Additionally, multilingual alert summaries in Sinhala and Tamil were automatically generated to improve communication accessibility in affected regions, ensuring that even non-English-speaking users could receive life-saving information in real time.

IV. RESULTS AND DISCUSSION

V. CONCLUTION

The rising frequency and severity of floods in Sri Lanka necessitates a more sophisticated, dependable, and rapid approach to flood forecasting and disaster response. This study introduces a novel Catchment-Aware Flood Forecasting System that combines IoT, Edge Computing, and Explainable AI (XAI) to address the critical limitations of existing flood monitoring systems, particularly data transmission delays and a lack of transparency in predictive modeling.

The system's utilization of edge computing allows for localized, real-time analysis of environmental data, which lowers latency and enhances early warning capabilities. The use of XAI guarantees that flood forecasts are not only precise but also comprehensible, giving emergency personnel, decision-makers, and communities at risk the knowledge they need to take prompt and efficient action.

By providing a mobile and web-based interface for real-time updates, route recommendations, and access to adjacent safe zones and necessary supplies, the suggested system goes beyond traditional alarm systems. This all-encompassing strategy encourages greater readiness, builds confidence in automated forecasts, and eventually strengthens community resilience.
In conclusion, the effective deployment of this system could revolutionize Sri Lanka's approach to flood risk management. It lays the foundation for an intelligent disaster management infrastructure at the national level by providing a flexible and scalable approach that can be expanded to other flood-prone areas. To promote long-term sustainability and adoption, future research will concentrate on improving system scalability, adding more environmental factors, and investigating commercialization prospects.

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